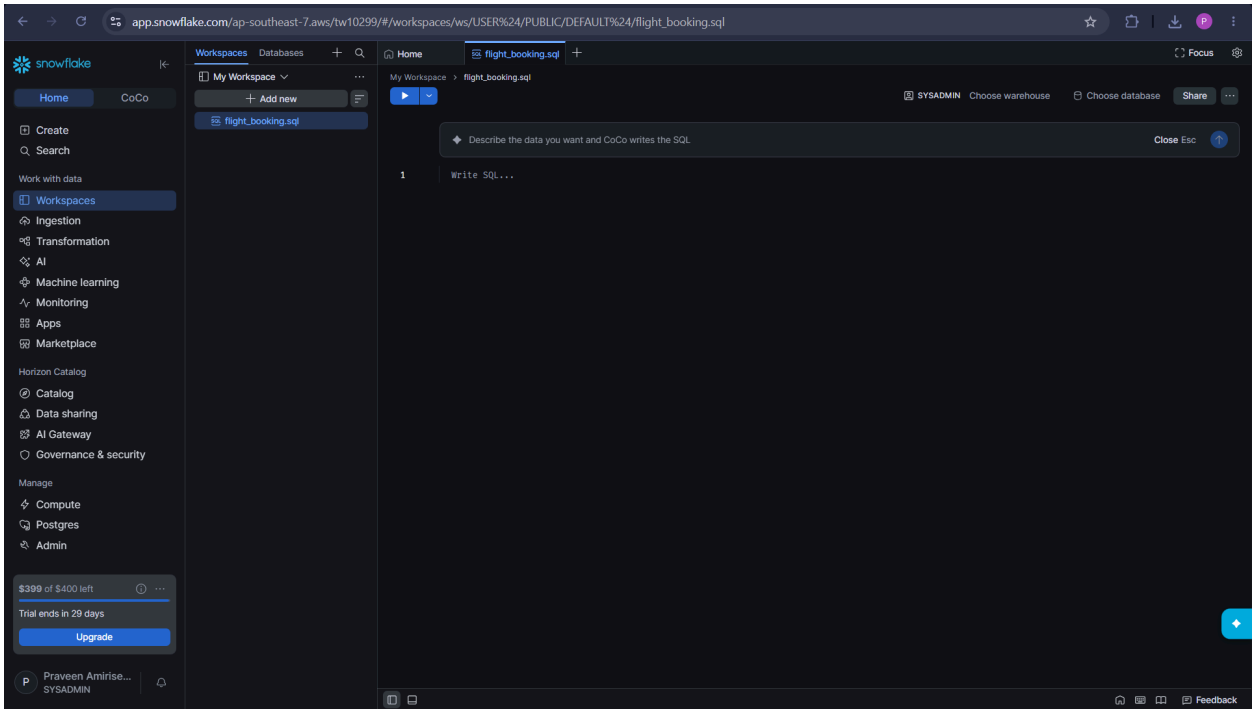
FLIGHT BOOKING MANAGEMENT



1. Database and schema

Create database

Command:

```
CREATE DATABASE FLIGHT_BOOKING_DB;
```

Output:

Results (1 minute ago)

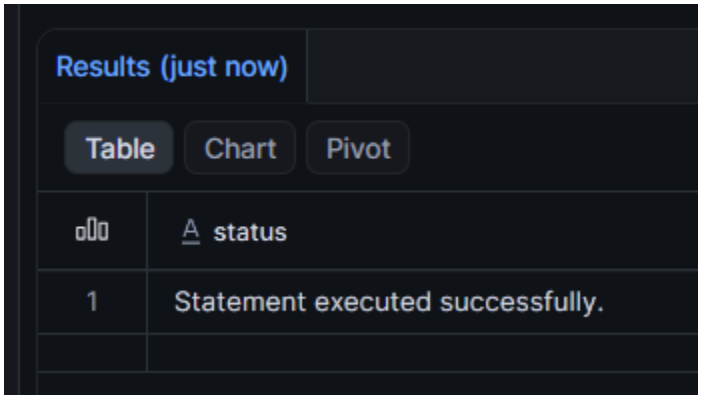
Table | Chart | Pivot

	status
1	Database FLIGHT_BOOKING_DB successfully created.

Command:

```
USE DATABASE FLIGHT_BOOKING_DB;
```

Output:

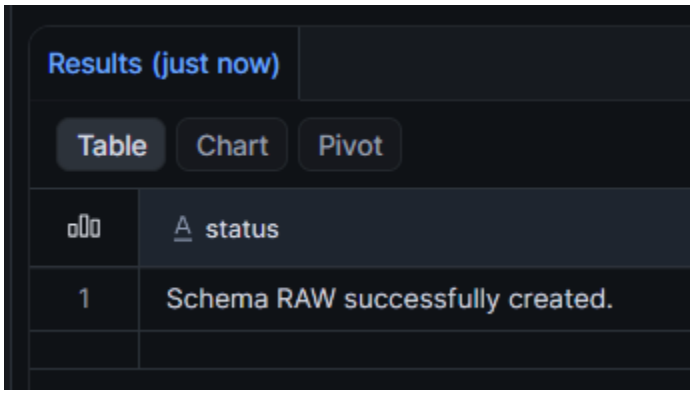


Create schema

Command:

```
CREATE SCHEMA RAW;
```

Output:



Command:

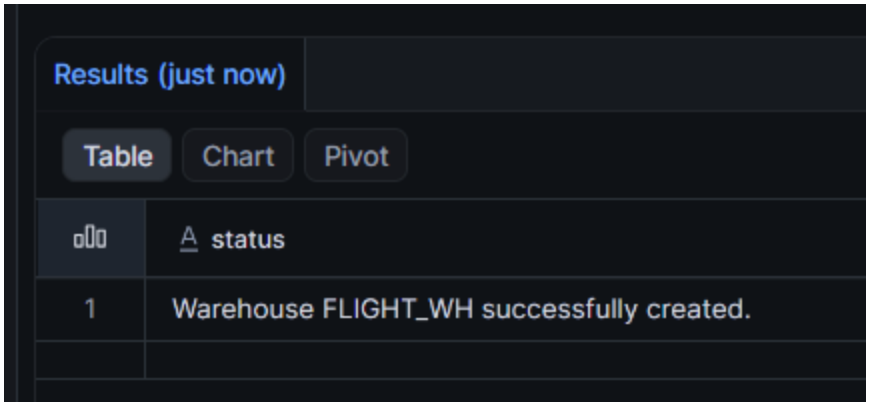
```
USE SCHEMA RAW;
```

Create warehouse

Command:

```
CREATE WAREHOUSE FLIGHT_WH
WAREHOUSE_SIZE = 'XSMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE;
```

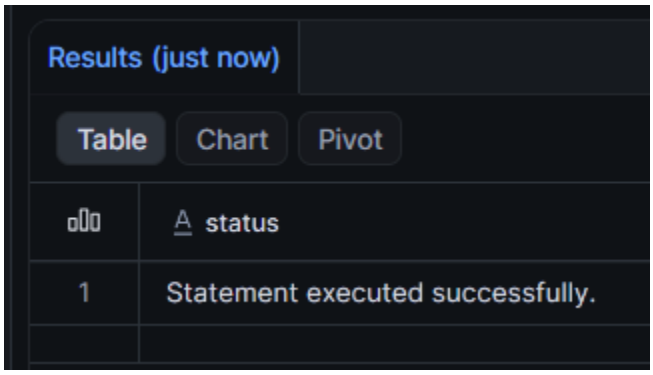
Output:



Command:

```
USE WAREHOUSE FLIGHT_WH;
```

Output:

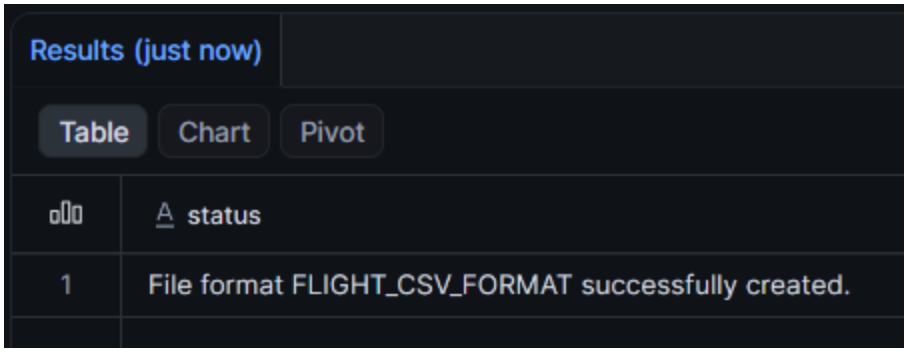


Create file format

Command:

```
CREATE OR REPLACE FILE FORMAT FLIGHT_CSV_FORMAT
TYPE = CSV
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
EMPTY_FIELD_AS_NULL = TRUE;
```

Output:

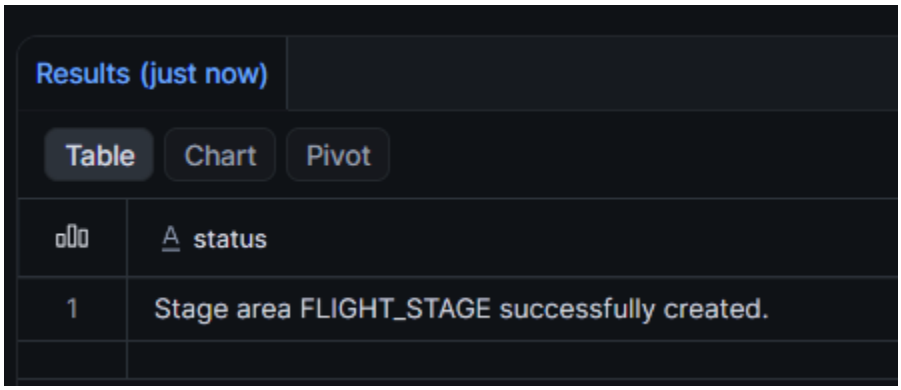


Create internal stage

Command:

```
CREATE STAGE FLIGHT_STAGE
FILE_FORMAT = FLIGHT_CSV_FORMAT;
```

Output:

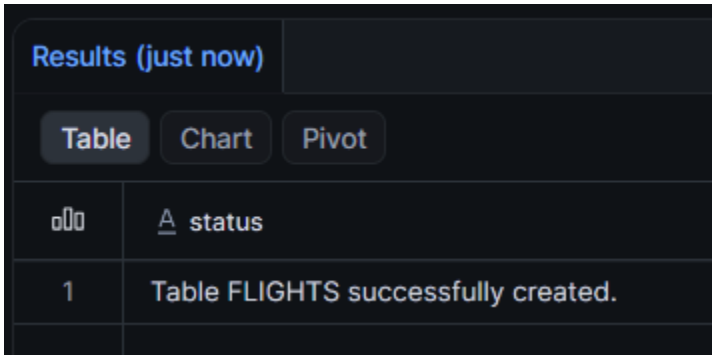


Create tables

Command:

```
CREATE OR REPLACE TABLE FLIGHTS (
  FLIGHT_ID VARCHAR(10),
  AIRLINE VARCHAR(60),
  SOURCE_CITY VARCHAR(60),
  DESTINATION_CITY VARCHAR(60),
  FLIGHT_DATE DATE,
  DEPARTURE_TIME TIME,
  ARRIVAL_TIME TIME,
  SEAT_CAPACITY NUMBER(5,0),
  BASE_FARE NUMBER(10,2)
);
```

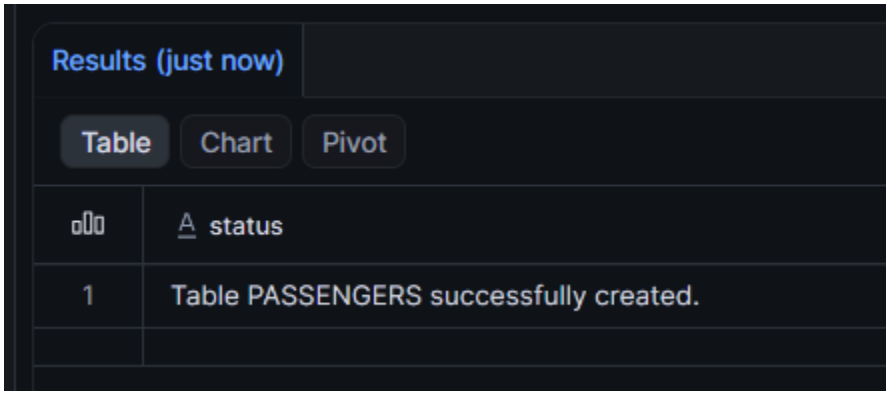
Output:



Command:

```
CREATE OR REPLACE TABLE PASSENGERS (
  PASSENGER_ID VARCHAR(10),
  PASSENGER_NAME VARCHAR(100),
  EMAIL VARCHAR(150),
  COUNTRY VARCHAR(50),
  DATE_OF_BIRTH DATE
);
```

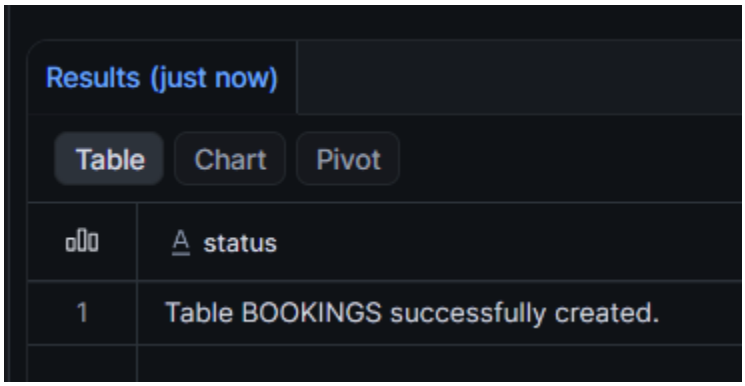
Output:



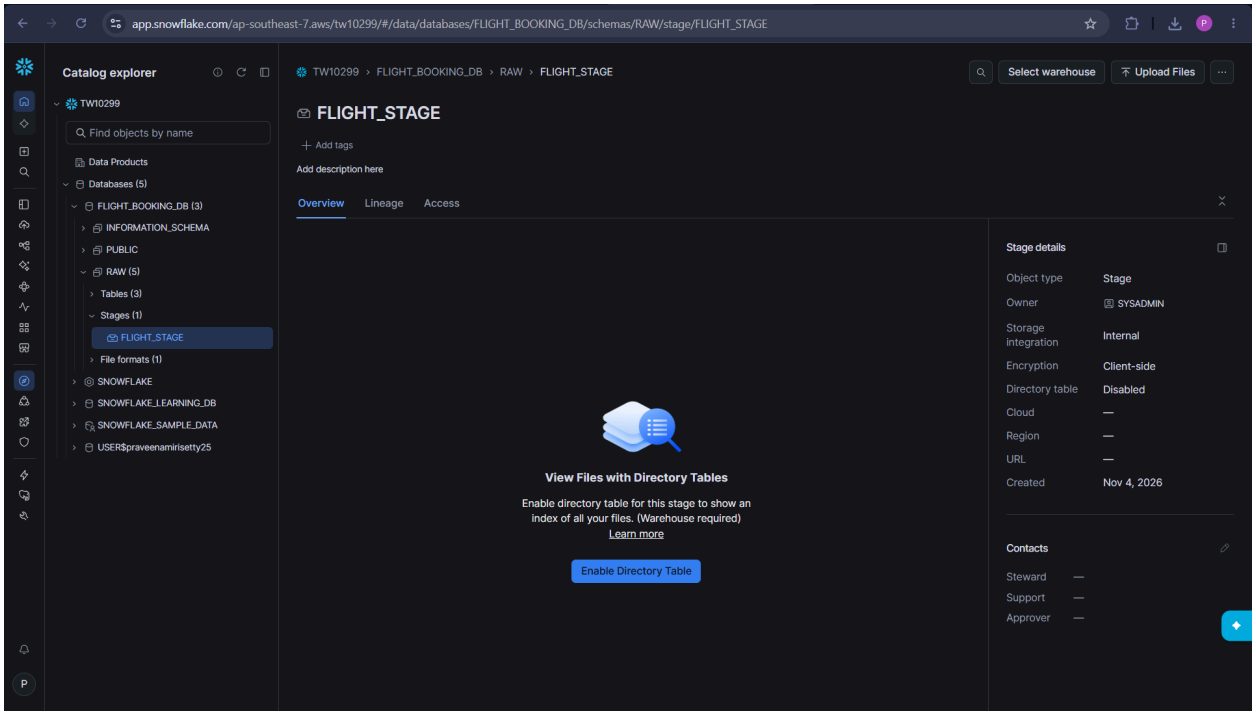
Command:

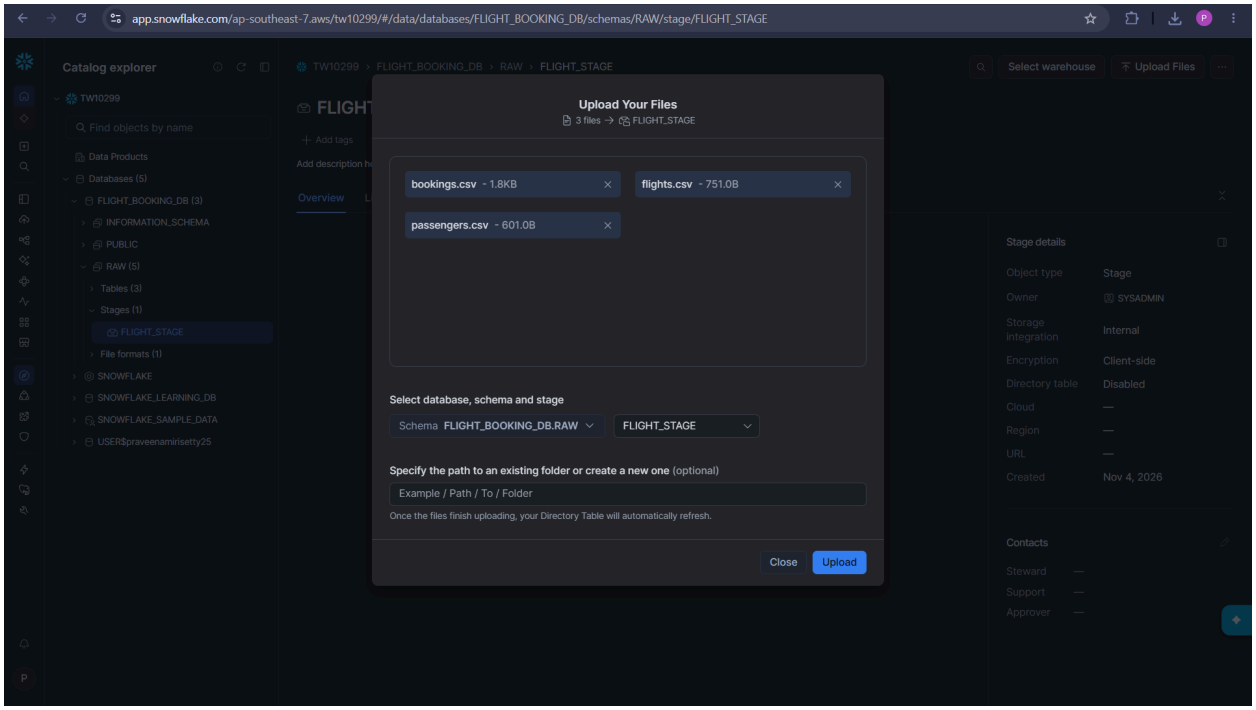
```sql
CREATE OR REPLACE TABLE BOOKINGS (
    BOOKING_ID VARCHAR(10),
    FLIGHT_ID VARCHAR(10),
    PASSENGER_ID VARCHAR(10),
    BOOKING_DATE DATE,
    TRAVEL_DATE DATE,
    SEAT_NUMBER VARCHAR(10),
    TICKET_FARE NUMBER(10,2),
    BOOKING_STATUS VARCHAR(20)
);
```

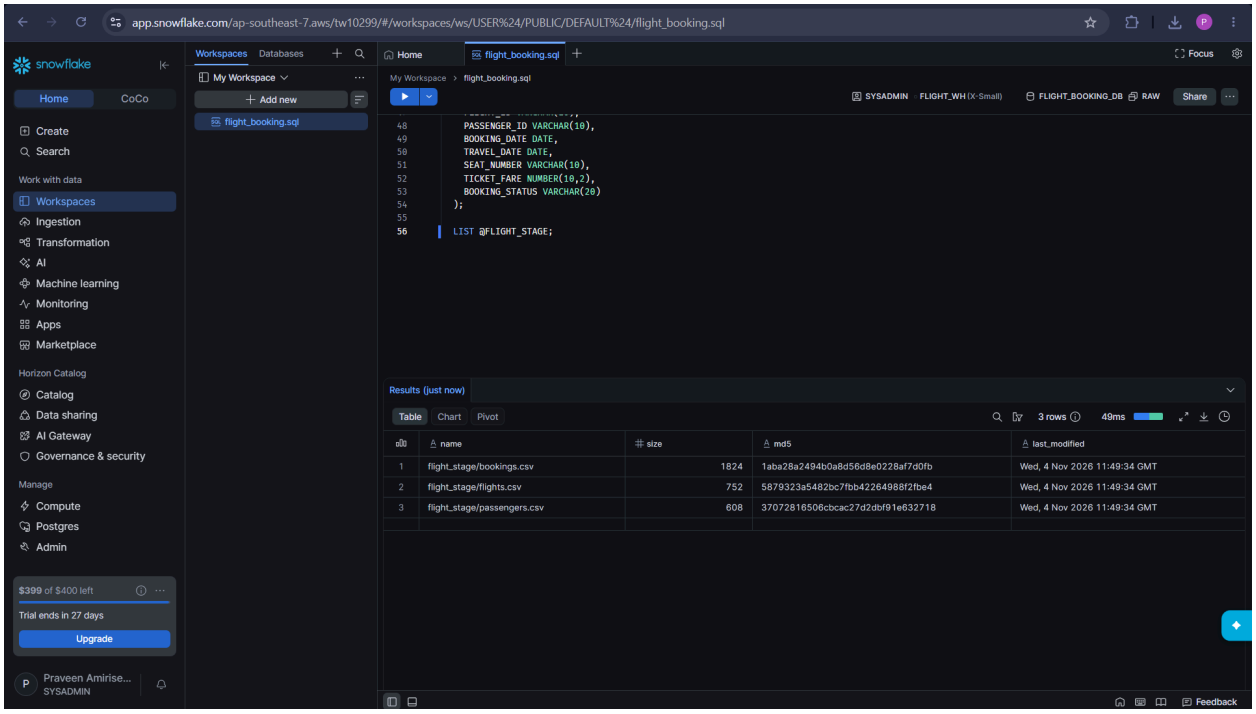
Output:



Upload datasets




app.snowflake.com/ap-southeast-7.aws/tw10299/#/data/databases/FLIGHT_BOOKING_DB/schemas/RAW/stage/FLIGHT_STAGE
Catalog explorer
TW10299 › FLIGHT_BOOKING_DB › RAW › FLIGHT_STAGE
Select warehouse
Upload Files
Upload Your Files
3 files → FLIGHT_STAGE
bookings.csv - 1.8KB
flights.csv - 751.0B
passengers.csv - 601.0B
Select database, schema and stage
Schema FLIGHT_BOOKING_DB.RAW
FLIGHT_STAGE
Specify the path to an existing folder or create a new one (optional)
Example / Path / To / Folder
Once the files finish uploading, your Directory Table will automatically refresh.
Close
Upload
Stage details
Object type Stage
Owner SYSADMIN
Storage integration Internal
Encryption Client-side
Directory table Disabled
Cloud —
Region —
URL —
Created Nov 4, 2026
Contacts
Steward —
Support —
Approver —


app.snowflake.com/ap-southeast-7.aws/tw10299/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/flight_booking.sql
snowflake
Workspaces
Databases
My Workspace
Add new
flight_booking.sql
Home
CoCo
Create
Search
Work with data
Workspaces
Ingestion
Transformation
AI
Machine learning
Monitoring
Apps
Marketplace
Horizon Catalog
Catalog
Data sharing
AI Gateway
Governance & security
Manage
Compute
Postgres
Admin
$399 of $400 left
Trial ends in 27 days
Upgrade
Praveen Amirise...
SYSADMIN
Home
flight_booking.sql
Focus
My Workspace › flight_booking.sql
SYSADMIN
FLIGHT_WH (X-Small)
FLIGHT_BOOKING_DB
RAW
Share
48 PASSENGER_ID VARCHAR(10),
49 BOOKING_DATE DATE,
50 TRAVEL_DATE DATE,
51 SEAT_NUMBER VARCHAR(10),
52 TICKET_FARE NUMBER(10,2),
53 BOOKING_STATUS VARCHAR(20)
54);
55
56 LIST @FLIGHT_STAGE;
Results (just now)
Table
Chart
Pivot
3 rows
49ms
name
size
md5
last_modified
1 flight_stage/bookings.csv 1824 1aba28a2494b0a8d56d8e0228af7d0fb Wed, 4 Nov 2026 11:49:34 GMT
2 flight_stage/flights.csv 752 5879323a5482bc7fbb4226498 8f2fbe4 Wed, 4 Nov 2026 11:49:34 GMT
3 flight_stage/passengers.csv 608 37072816506cbcac27d2dbf91e632718 Wed, 4 Nov 2026 11:49:34 GMT
Feedback

Load CSV data

Command:

```
COPY INTO FLIGHTS
FROM @FLIGHT_STAGE/flights.csv
FILE_FORMAT = (FORMAT_NAME = FLIGHT_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';
```

Output:

Results (just now)

Pivot 1 row 738ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line
1	flight_stage/flights.csv	LOADED	10	10	1	0	null	null

Command:

```
COPY INTO PASSENGERS
FROM @FLIGHT_STAGE/passengers.csv
FILE_FORMAT = (FORMAT_NAME = FLIGHT_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';
```

Output:

Results (just now)

Table Chart Pivot 1 row 405ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line
1	flight_stage/passenger	LOADED	10	10	1	0	null	null

Command:

```
COPY INTO BOOKINGS
FROM @FLIGHT_STAGE/bookings.csv
FILE_FORMAT = (FORMAT_NAME = FLIGHT_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';
```

Output:

Results (just now)

Table Chart Pivot 1 row 470ms

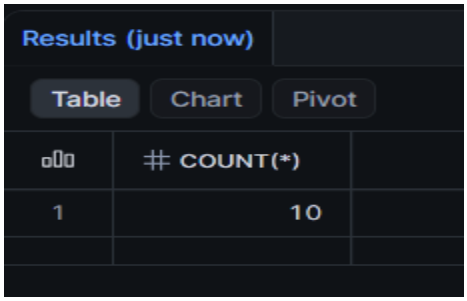
	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line
1	flight_stage/bookings.c	LOADED	30	30	1	0	null	null

Verify records

Command:

```
SELECT COUNT(*) FROM FLIGHTS;
```

Output:



Command:

```
SELECT COUNT(*) FROM PASSENGERS;
```

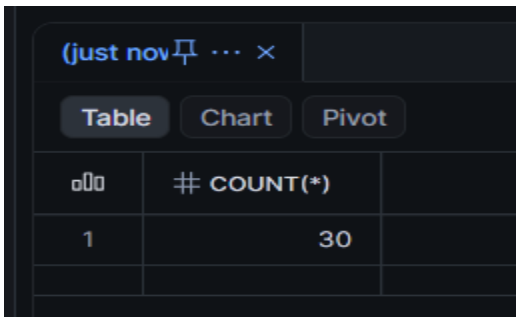
Output:



Command:

```
SELECT COUNT(*) FROM BOOKINGS;
```

Output:

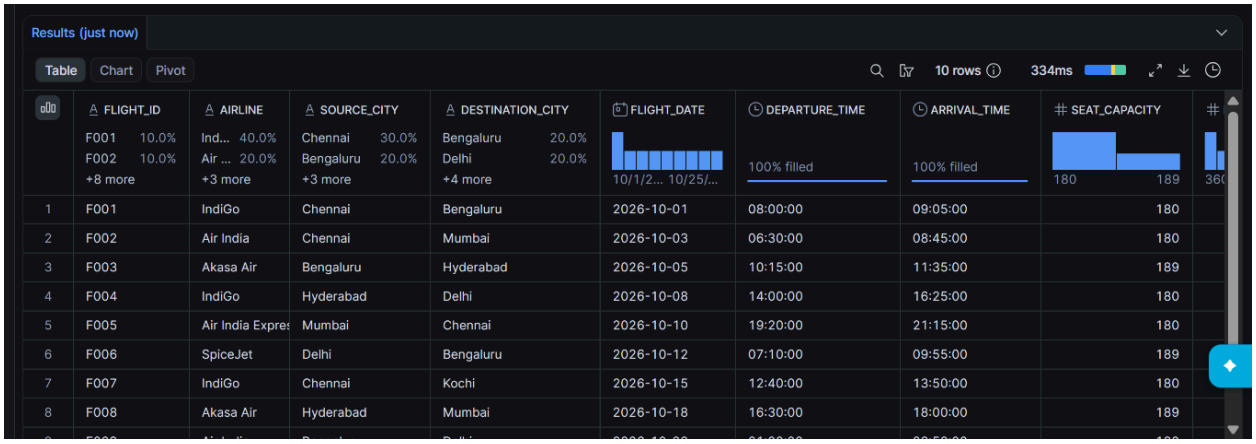


SQL analysis

Command:

```
SELECT * FROM FLIGHTS ORDER BY FLIGHT_ID;
```

Output:



Results (just now)

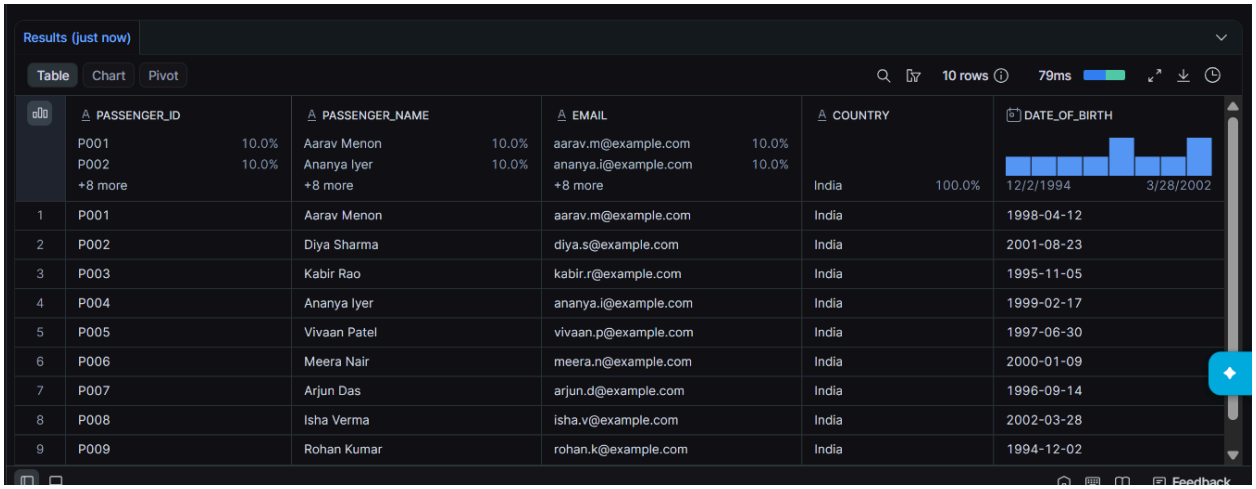
Table | Chart | Pivot — 10 rows — 334ms

	FLIGHT_ID	AIRLINE	SOURCE_CITY	DESTINATION_CITY	FLIGHT_DATE	DEPARTURE_TIME	ARRIVAL_TIME	SEAT_CAPACITY
	F001 10.0%, F002 10.0%, +8 more	Ind... 40.0%, Air ... 20.0%, +3 more	Chennai 30.0%, Bengaluru 20.0%, +3 more	Bengaluru 20.0%, Delhi 20.0%, +4 more	10/1/2... 10/25/...	100% filled	100% filled	180 189
1	F001	IndiGo	Chennai	Bengaluru	2026-10-01	08:00:00	09:05:00	180
2	F002	Air India	Chennai	Mumbai	2026-10-03	06:30:00	08:45:00	180
3	F003	Akasa Air	Bengaluru	Hyderabad	2026-10-05	10:15:00	11:35:00	189
4	F004	IndiGo	Hyderabad	Delhi	2026-10-08	14:00:00	16:25:00	180
5	F005	Air India Expres	Mumbai	Chennai	2026-10-10	19:20:00	21:15:00	180
6	F006	SpiceJet	Delhi	Bengaluru	2026-10-12	07:10:00	09:55:00	189
7	F007	IndiGo	Chennai	Kochi	2026-10-15	12:40:00	13:50:00	180
8	F008	Akasa Air	Hyderabad	Mumbai	2026-10-18	16:30:00	18:00:00	189

Command:

```
SELECT * FROM PASSENGERS ORDER BY PASSENGER_ID;
```

Output:



Results (just now)

Table | Chart | Pivot — 10 rows — 79ms

	PASSENGER_ID	PASSENGER_NAME	EMAIL	COUNTRY	DATE_OF_BIRTH
	P001 10.0%, P002 10.0%, +8 more	Aarav Menon 10.0%, Ananya Iyer 10.0%, +8 more	aarav.m@example.com 10.0%, ananya.i@example.com 10.0%, +8 more	India 100.0%	12/2/1994 3/28/2002
1	P001	Aarav Menon	aarav.m@example.com	India	1998-04-12
2	P002	Diya Sharma	diya.s@example.com	India	2001-08-23
3	P003	Kabir Rao	kabir.r@example.com	India	1995-11-05
4	P004	Ananya Iyer	ananya.i@example.com	India	1999-02-17
5	P005	Vivaan Patel	vivaan.p@example.com	India	1997-06-30
6	P006	Meera Nair	meera.n@example.com	India	2000-01-09
7	P007	Arjun Das	arjun.d@example.com	India	1996-09-14
8	P008	Isha Verma	isha.v@example.com	India	2002-03-28
9	P009	Rohan Kumar	rohan.k@example.com	India	1994-12-02

Feedback

Command:

```
SELECT * FROM BOOKINGS ORDER BY BOOKING_ID;
```

Output:

Results (just now)

Table | Chart | Pivot — 30 rows — 132ms

	BOOKING_ID	FLIGHT_ID	PASSENGER_ID	BOOKING_DATE	TRAVEL_DATE	SEAT_NUMBER	TICKET_FARE	BOOKING_STATUS
	B001 3.3% B002 3.3% +28 more	F001 10.0% F002 10.0% +8 more	P001 10.0% P002 10.0% +8 more	9/10/20... 10/6/20...	10/1/2... 10/25/...	14C 6.7% 06A 3.3% +27 more	3600 8100	COMPLETED 66.7% CANCELLED 20.0% +1 more
1	B001	F001	P001	2026-09-10	2026-10-01	12A	4200.00	COMPLETED
2	B002	F001	P002	2026-09-12	2026-10-01	12B	4200.00	COMPLETED
3	B003	F001	P003	2026-09-15	2026-10-01	14C	4400.00	CANCELLED
4	B004	F002	P004	2026-09-11	2026-10-03	08A	6800.00	COMPLETED
5	B005	F002	P005	2026-09-13	2026-10-03	08B	7000.00	COMPLETED
6	B006	F002	P006	2026-09-18	2026-10-03	09C	6800.00	CANCELLED
7	B007	F003	P007	2026-09-16	2026-10-05	15A	3900.00	COMPLETED
8	B008	F003	P008	2026-09-17	2026-10-05	15B	4100.00	COMPLETED
9	B009	F003	P009	2026-09-20	2026-10-05	16C	3900.00	CONFIRMED

Command:

```
SELECT AIRLINE, COUNT(*) AS NUMBER_OF_FLIGHTS
FROM FLIGHTS GROUP BY AIRLINE ORDER BY NUMBER_OF_FLIGHTS DESC;
```

Output:

Results (just now)

Table | Chart | Pivot — 5 rows — 72ms

	AIRLINE	NUMBER_OF_FLIGHTS
1	IndiGo	4
2	Air India	2
3	Akasa Air	2
4	Air India Express	1
5	SpiceJet	1

Command:

```
SELECT BOOKING_STATUS, COUNT(*) AS NUMBER_OF_BOOKINGS
FROM BOOKINGS GROUP BY BOOKING_STATUS ORDER BY NUMBER_OF_BOOKINGS DESC;
```

Output:

Results (just now)

Table | Chart | Pivot — 3 rows — 73ms

	BOOKING_STATUS	NUMBER_OF_BOOKINGS
1	COMPLETED	20
2	CANCELLED	6
3	CONFIRMED	4

Command:

```
SELECT SOURCE_CITY, DESTINATION_CITY, COUNT(*) AS ROUTE_COUNT
FROM FLIGHTS GROUP BY SOURCE_CITY, DESTINATION_CITY;
```

Output:



Results (just now)

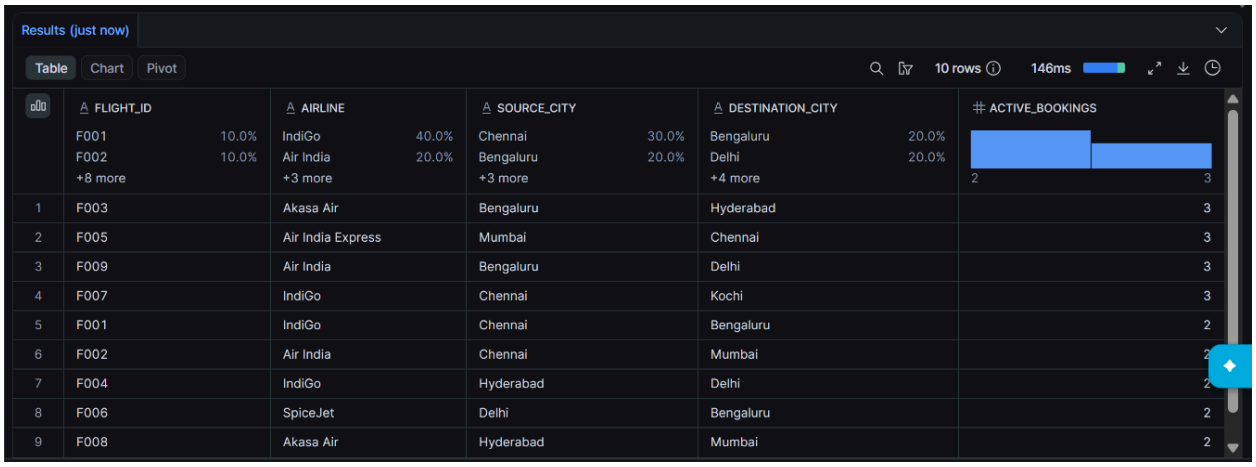
Table | Chart | Pivot — 10 rows — 57ms

	SOURCE_CITY	DESTINATION_CITY	ROUTE_COUNT
	Chennai 30.0% Bengaluru 20.0% +3 more	Bengaluru 20.0% Delhi 20.0% +4 more	1 … 1
1	Chennai	Bengaluru	1
2	Chennai	Mumbai	1
3	Bengaluru	Hyderabad	1
4	Hyderabad	Delhi	1
5	Delhi	Bengaluru	1
6	Chennai	Kochi	1
7	Hyderabad	Mumbai	1
8	Mumbai	Hyderabad	1
9	Bengaluru	Delhi	1

Command: Confirmed/completed bookings by route and flight

```
SELECT F.FLIGHT_ID, F.AIRLINE, F.SOURCE_CITY, F.DESTINATION_CITY,
       COUNT_IF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED')) AS ACTIVE_BOOKINGS
FROM FLIGHTS F LEFT JOIN BOOKINGS B ON F.FLIGHT_ID = B.FLIGHT_ID
GROUP BY F.FLIGHT_ID, F.AIRLINE, F.SOURCE_CITY, F.DESTINATION_CITY
ORDER BY ACTIVE_BOOKINGS DESC;
```

Output:



Results (just now)

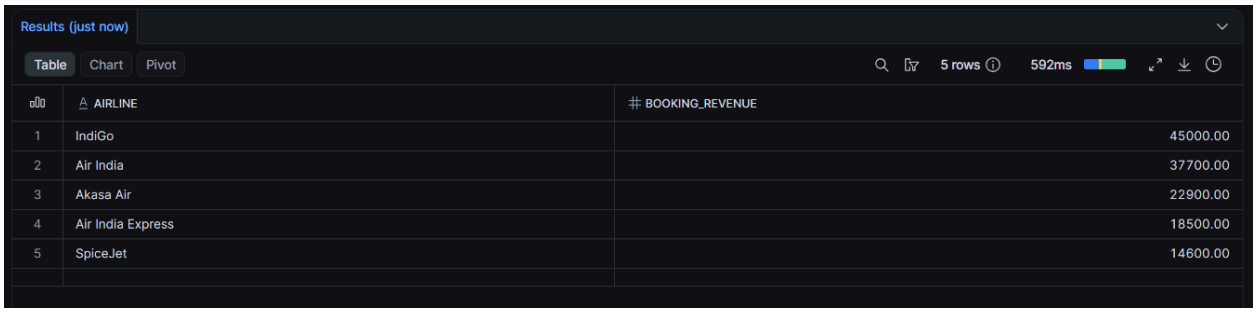
Table | Chart | Pivot — 10 rows — 146ms

	FLIGHT_ID	AIRLINE	SOURCE_CITY	DESTINATION_CITY	ACTIVE_BOOKINGS
	F001 10.0% F002 10.0% +8 more	IndiGo 40.0% Air India 20.0% +3 more	Chennai 30.0% Bengaluru 20.0% +3 more	Bengaluru 20.0% Delhi 20.0% +4 more	2 … 3
1	F003	Akasa Air	Bengaluru	Hyderabad	3
2	F005	Air India Express	Mumbai	Chennai	3
3	F009	Air India	Bengaluru	Delhi	3
4	F007	IndiGo	Chennai	Kochi	3
5	F001	IndiGo	Chennai	Bengaluru	2
6	F002	Air India	Chennai	Mumbai	2
7	F004	IndiGo	Hyderabad	Delhi	2
8	F006	SpiceJet	Delhi	Bengaluru	2
9	F008	Akasa Air	Hyderabad	Mumbai	2

Command: Revenue by airline (exclude cancelled bookings)

```
SELECT F.AIRLINE,
       SUM(IFF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED'), B.TICKET_FARE, 0)) AS BOOKING_REVENUE
FROM FLIGHTS F JOIN BOOKINGS B ON F.FLIGHT_ID = B.FLIGHT_ID
GROUP BY F.AIRLINE ORDER BY BOOKING_REVENUE DESC;
```

Output:



Results (just now)

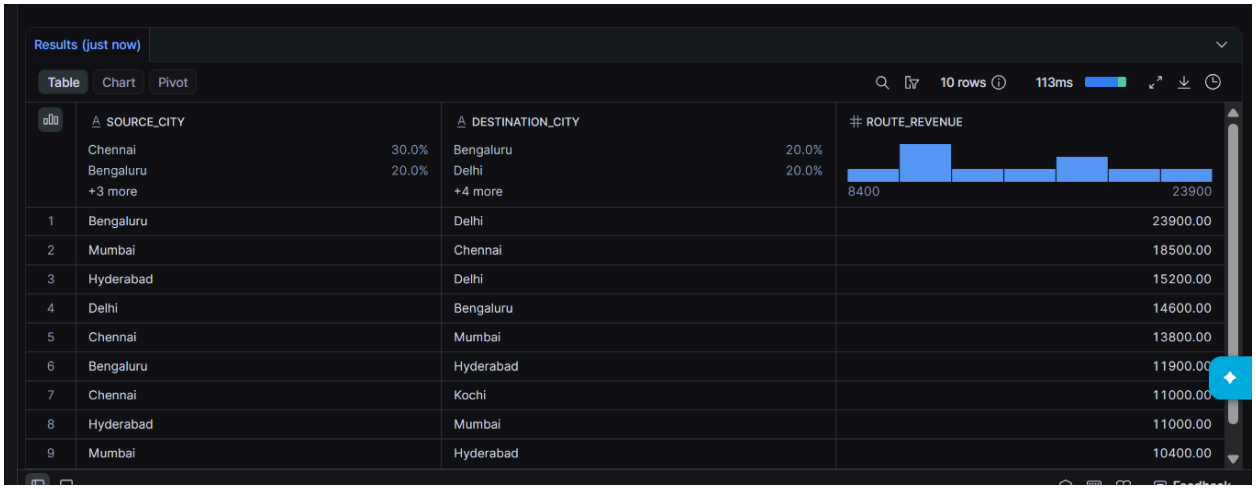
Table | Chart | Pivot — 5 rows — 592ms

	AIRLINE	BOOKING_REVENUE
1	IndiGo	45000.00
2	Air India	37700.00
3	Akasa Air	22900.00
4	Air India Express	18500.00
5	SpiceJet	14600.00

Command: Highest revenue routes (exclude cancelled bookings)

```
SELECT F.SOURCE_CITY, F.DESTINATION_CITY,
       SUM(IFF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED'), B.TICKET_FARE, 0)) AS ROUTE_REVENUE
FROM FLIGHTS F JOIN BOOKINGS B ON F.FLIGHT_ID = B.FLIGHT_ID
GROUP BY F.SOURCE_CITY, F.DESTINATION_CITY
ORDER BY ROUTE_REVENUE DESC;
```

Output:



Results (just now)

Table | Chart | Pivot — 10 rows — 113ms

	SOURCE_CITY	DESTINATION_CITY	ROUTE_REVENUE
	Chennai 30.0% Bengaluru 20.0% +3 more	Bengaluru 20.0% Delhi 20.0% +4 more	8400 – 23900
1	Bengaluru	Delhi	23900.00
2	Mumbai	Chennai	18500.00
3	Hyderabad	Delhi	15200.00
4	Delhi	Bengaluru	14600.00
5	Chennai	Mumbai	13800.00
6	Bengaluru	Hyderabad	11900.00
7	Chennai	Kochi	11000.00
8	Hyderabad	Mumbai	11000.00
9	Mumbai	Hyderabad	10400.00

Command: Passenger booking history

```
SELECT P.PASSENGER_ID, P.PASSENGER_NAME, B.BOOKING_ID,
       F.AIRLINE, F.SOURCE_CITY, F.DESTINATION_CITY,
       B.TRAVEL_DATE, B.TICKET_FARE, B.BOOKING_STATUS
FROM PASSENGERS P
JOIN BOOKINGS B ON P.PASSENGER_ID = B.PASSENGER_ID
JOIN FLIGHTS F ON B.FLIGHT_ID = F.FLIGHT_ID
ORDER BY P.PASSENGER_ID, B.TRAVEL_DATE;
```

Output:



Results (just now)

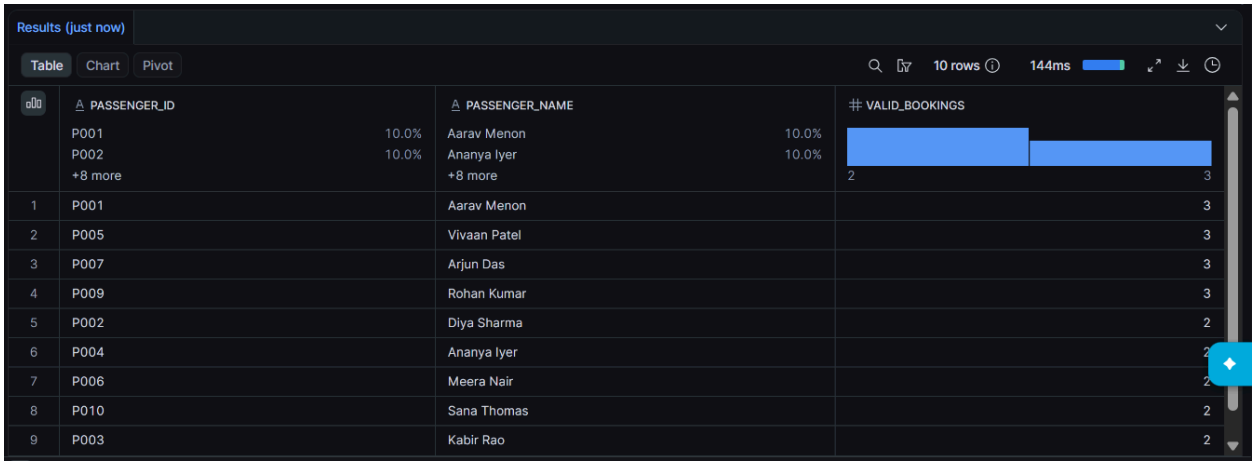
Table | Chart | Pivot — 30 rows — 128ms

	PASSENGER_ID	PASSENGER_NAME	BOOKING_ID	AIRLINE	SOURCE_CITY	DESTINATION_CITY	TRAVEL_DATE	TICKET_FARE
	P001 10.0% P002 10.0% +8 more	Aarav Menon 10.0% Ananya Iyer 10.0% +8 more	B001 3.3% B002 3.3% +28 more	Ind... 40.0% Air ... 20.0% +3 more	Chennai 30.0% Bengaluru 20.0% +3 more	Bengaluru 20.0% Delhi 20.0% +4 more	10/1/2... 10/25/...	3600 8100
1	P001	Aarav Menon	B001	IndiGo	Chennai	Bengaluru	2026-10-01	4200.00
2	P001	Aarav Menon	B011	IndiGo	Hyderabad	Delhi	2026-10-08	7700.00
3	P001	Aarav Menon	B021	IndiGo	Chennai	Kochi	2026-10-15	3600.00
4	P002	Diya Sharma	B002	IndiGo	Chennai	Bengaluru	2026-10-01	4200.00
5	P002	Diya Sharma	B012	IndiGo	Hyderabad	Delhi	2026-10-08	7500.00
6	P002	Diya Sharma	B022	Akasa Air	Hyderabad	Mumbai	2026-10-18	5400.00
7	P003	Kabir Rao	B003	IndiGo	Chennai	Bengaluru	2026-10-01	4400.00
8	P003	Kabir Rao	B013	Air India Expres	Mumbai	Chennai	2026-10-10	6100.00

Command: Top passengers by number of non-cancelled bookings

```
SELECT P.PASSENGER_ID, P.PASSENGER_NAME,
       COUNT_IF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED')) AS VALID_BOOKINGS
FROM PASSENGERS P JOIN BOOKINGS B ON P.PASSENGER_ID = B.PASSENGER_ID
GROUP BY P.PASSENGER_ID, P.PASSENGER_NAME
ORDER BY VALID_BOOKINGS DESC;
```

Output:



Results (just now)

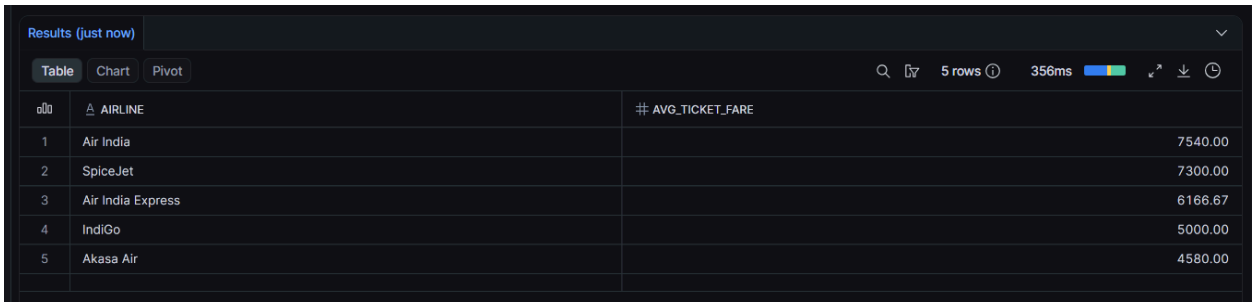
Table | Chart | Pivot — 10 rows — 144ms

	PASSENGER_ID	PASSENGER_NAME	VALID_BOOKINGS
	P001 10.0% P002 10.0% +8 more	Aarav Menon 10.0% Ananya Iyer 10.0% +8 more	2 3
1	P001	Aarav Menon	3
2	P005	Vivaan Patel	3
3	P007	Arjun Das	3
4	P009	Rohan Kumar	3
5	P002	Diya Sharma	2
6	P004	Ananya Iyer	2
7	P006	Meera Nair	2
8	P010	Sana Thomas	2
9	P003	Kabir Rao	2

Command: Average ticket fare by airline (non-cancelled bookings)

```
SELECT F.AIRLINE, ROUND(AVG(B.TICKET_FARE),2) AS AVG_TICKET_FARE
FROM FLIGHTS F JOIN BOOKINGS B ON F.FLIGHT_ID = B.FLIGHT_ID
WHERE B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED')
GROUP BY F.AIRLINE ORDER BY AVG_TICKET_FARE DESC;
```

Output:



Results (just now)

Table | Chart | Pivot — 5 rows — 356ms

	AIRLINE	AVG_TICKET_FARE
1	Air India	7540.00
2	SpiceJet	7300.00
3	Air India Express	6166.67
4	IndiGo	5000.00
5	Akasa Air	4580.00

Command: Flight occupancy percentage based on non-cancelled bookings

```
SELECT F.FLIGHT_ID, F.AIRLINE, F.SEAT_CAPACITY,
       COUNT_IF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED')) AS BOOKED_SEATS,
       ROUND(100.0 * COUNT_IF(B.BOOKING_STATUS IN ('CONFIRMED','COMPLETED')) / NULLIF(F.SEAT_CAPACITY,0),2) AS OCCUPANCY_PERCENT
FROM FLIGHTS F LEFT JOIN BOOKINGS B ON F.FLIGHT_ID = B.FLIGHT_ID
GROUP BY F.FLIGHT_ID, F.AIRLINE, F.SEAT_CAPACITY
ORDER BY OCCUPANCY_PERCENT DESC;
```

Output:



Results (just now)

Table | Chart | Pivot — 10 rows — 102ms

	FLIGHT_ID	AIRLINE	SEAT_CAPACITY	BOOKED_SEATS	OCCUPANCY_PERCENT
	F001 10.0% F002 10.0% +8 more	IndiGo 40.0% Air India 20.0% +3 more	180 … 189	2 … 3	1.06 … 1.67
1	F007	IndiGo	180	3	1.67
2	F009	Air India	180	3	1.67
3	F005	Air India Express	180	3	1.67
4	F003	Akasa Air	189	3	1.59
5	F001	IndiGo	180	2	1.11
6	F002	Air India	180	2	1.11
7	F004	IndiGo	180	2	1.11
8	F010	IndiGo	180	2	1.11
9	F006	SpiceJet	189	2	1.06

Command:

```
SELECT
  TRIM('   Snowflake Flight Project   ') AS TRIMMED_TEXT,
  TRIM('   ') AS TRIMMED_WHITESPACE,
  NULLIF(TRIM('   '), '') AS WHITESPACE_TO_NULL,
  NULLIF(TRIM('  Chennai  '), '') AS CLEAN_CITY,
  NULLIF(TRIM(''), '') AS EMPTY_TO_NULL;
```

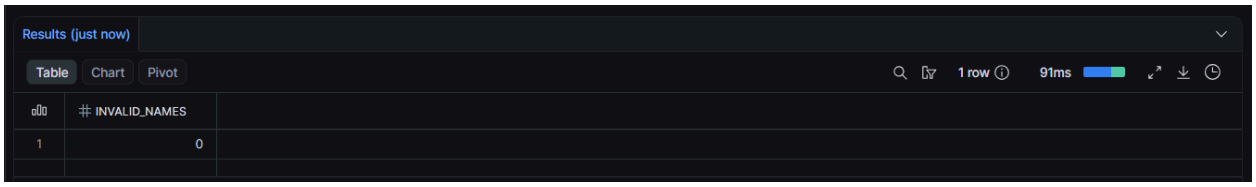
Output:



Command:

```
SELECT COUNT_IF(PASSENGER_NAME IS NULL OR TRIM(PASSENGER_NAME) = '') AS INVALID_NAMES
FROM PASSENGERS;
```

Output:

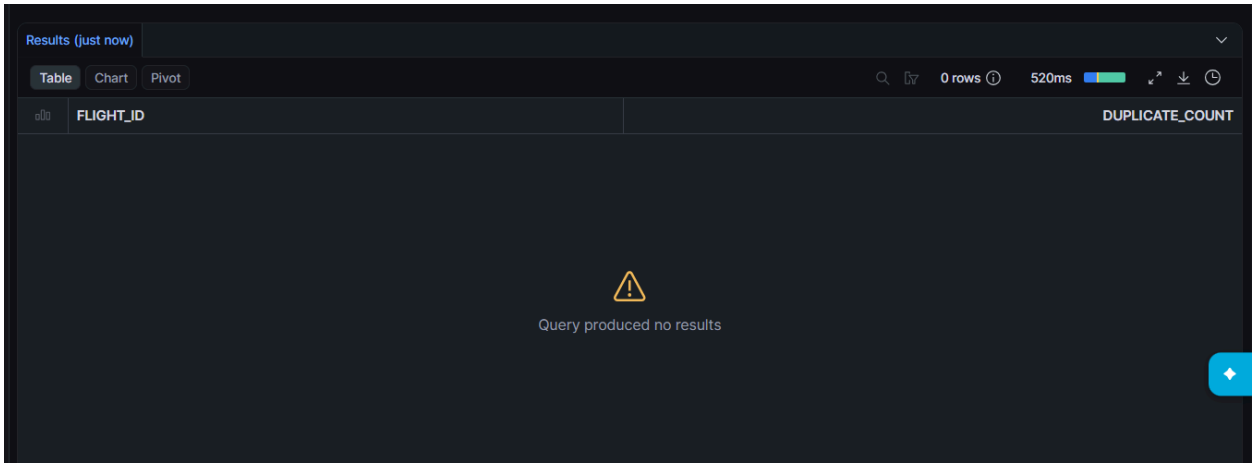


Data-quality checks

Command: Duplicate primary identifiers

```
SELECT FLIGHT_ID, COUNT(*) AS DUPLICATE_COUNT FROM FLIGHTS
GROUP BY FLIGHT_ID HAVING COUNT(*) > 1;
```

Output:



Command: Date and fare checks

```
SELECT * FROM BOOKINGS
WHERE TRAVEL_DATE < BOOKING_DATE OR TICKET_FARE < 0;
```

Output:

